Requirements Traceability Matrix (RTM): RAG-Diary

> A trace table that links each requirement back to the gap found in the as-is analysis, and forward to its use case, user story, real screen, real API endpoint and test case.

> Thai version: TRACEABILITY-MATRIX.md

1. Document control

Item	Detail
Document	Requirements Traceability Matrix
Version	1.0
Prepared	15 October 2026
Author	Business Analyst / System Analyst (project owner)
Status	Reviewed, round 1 (25 October 2026)
Related documents	SRS, USER-STORIES, USE-CASES, TEST-PLAN

2. Purpose and how to read it

This document answers one question: what did each requirement become in the system, and is there a test point that covers it?

Reading left to right is forward tracing: gap, then requirement, then screen, then API, then test case. Reading right to left is backward tracing: from any test point you can see which requirements it covers.

A dash means there is no direct link.

3. Requirements traceability matrix

Requirement	AS-IS gap	Use case	User story	Real screen	Real API endpoint	Test case
FR-01 Register an account	GAP-05	UC-01	US-01	/register	POST /auth/register	TC-01
FR-02 Log in	GAP-05	UC-01	US-01	/login	POST /auth/login, POST /auth/refresh	TC-02

Requirement	AS-IS gap	Use case	User story	Real screen	Real API endpoint	Test case
FR-03 Log out	GAP-05	UC-01	US-01	All pages	POST /auth/logout	TC-03
FR-04 Write a new entry with a tag	GAP-06	UC-02	US-02	/diary	POST /notes	TC-04
FR-05 Build the semantic index automatically	GAP-01	UC-02	US-03	/diary	POST /notes (server side)	TC-05
FR-06 View one's own entries	GAP-03	UC-03	US-04	/ and /historyBoard	GET /notes	TC-06
FR-07 Edit an entry	GAP-06	UC-03	US-04	/historyBoard	PATCH /notes/:id	TC-07
FR-08 Delete a single entry	GAP-06	UC-03	US-04	/historyBoard	DELETE /notes/:id	TC-08
FR-09 Semantic search	GAP-01, GAP-02	UC-04	US-05	/search	POST /search/semantic	TC-09
FR-10 Chat with one's own entries	GAP-02	UC-04	US-06	/search	POST /chat	TC-10
FR-11 Mood summary as text and an image prompt	GAP-04	UC-05	US-07	/diary	POST /ai/summarize	TC-11
FR-12 Generate an illustration from the prompt	GAP-04	UC-05	US-08	/diary	POST /ai/imagine	TC-12, TC-24
FR-13 Export all data	GAP-05	UC-06	US-09	/setting	GET /export	TC-13
FR-14 Delete all data	GAP-05	UC-06	US-09	/setting	DELETE /data	TC-14
FR-15 Isolate data per user	GAP-05	All UCs	US-10	All pages	Every data endpoint	TC-15
NFR-01 Performance	GAP-02	UC-04	US-05	/search	POST /search/semantic	TC-16

Requirement	AS-IS gap	Use case	User story	Real screen	Real API endpoint	Test case
NFR-02 Security	GAP-05	UC-01, UC-06	US-10	All pages	Every data endpoint	TC-17
NFR-03 Privacy (PDPA)	GAP-05	UC-02, UC-04, UC-05, UC-06	US-03, US-05, US-06, US-07, US-09	/setting and the AI feature screens	POST /notes, POST /search/semantic, POST /chat, POST /ai/summarize, POST /ai/imagine	TC-18
NFR-04 Usability and accessibility	-	All UCs	All US	All pages	-	TC-19
NFR-05 Error handling	GAP-02	UC-02, UC-04, UC-05	US-06, US-08	/search, /diary	POST /chat, POST /ai/imagine	TC-20
NFR-06 Scalability	GAP-02	UC-04	US-05	/search	POST /search/semantic	TC-21
NFR-07 Maintainability	-	-	-	-	-	TC-22
NFR-08 Compatibility and responsive layout	-	All UCs	All US	All pages	-	TC-23

4. Coverage summary

Measure	Value
Total requirements	23
Requirements with a link in every relevant column	23
Coverage	100 percent
Requirements with no screen	0
Requirements with no test case	0

NFR-07 (maintainability) has no natural link to a use case, screen or API, because it is a property of the development process. It is checked by code review (TC-22).

5. Improvement points

The items below are not defects in the requirements. They are design gaps that should be closed in the next round.

ID	Topic	Proposal
FR-13	Closed: GET /export was added on the server and the settings page calls it	-
FR-14	Closed: DELETE /data deletes in one call and the settings page calls it	-

6. References

Document	Relationship
SRS	The origin of the requirement IDs
AS-IS-TO-BE	The origin of the GAP IDs
USER-STORIES	The origin of the user story IDs
USE-CASES	The origin of the UC IDs
TEST-PLAN	The origin of the TC IDs and the test detail